THREE PROOFS OF THE CATALAN CONJECTURE

SHANG-SHAN CHONG, PH.D

Abstract. Catalan's Conjecture was proven by Mihailescu in 2004. In this paper, I offer three fairly direct proofs. First, using calculus and geometric reasoning, the solution is shown to be unique. Secondly, the Catalan polynomial is explicitly constructed, and factored which assumes fix prime exponents. Using careful reasoning about orders of multiplicative prime fields, the condition $C < B$ is shown to yield no solutions whether C, B are even or odd, C prime or composite. And when $B < C$, careful reasoning about the factors in the Catalan polynomial shows that $3^2 - 2^3 = 1$ is unique, ignorin some trivial cases. And finally, a third attempt is made on a more general problem: $x^{p-1} \mid (p-1)^x + 1$, this time using elementary number theory not more complicated than Bezout's Theorem and Fermat's Little Theorem. The solutions found can be checked against the original Catalan equation to discover that $3^2 - 2^3 = 1$ is unique.

1. Introduction

The Catalan's Conjecture, solved by Mihăilescu [1], states that

Theorem (Catalan's Conjecture, solved by Mihăilescu)**.**

$$B^y = C^z - 1 \tag{1}$$

has no solution other than $2^3 = 3^2 - 1$*, when* $y, z > 1$*.*

2. Motivation

The reason for what I am doing is not to undercut Mihailescu's [1] achievement. After all, being first to solve a problem is no simple feat. His proof serves the purpose of linking disparate fields of pure mathematics into a larger whole, and that is the value of his achievement.

What I want is a simpler proof of the Catalan Conjecture than that described by Mihailescu, using less complex abstract algebraic machinery. The book [2] is of no help as it presents only a sketchy proof of the details. I believe my proof serves as an insightful look, getting to the essentials of a problem that had stymied generations of mathematicians, until Mihailescu.

Claim 1. *It may be assumed that* y *and* z *in equation* (1) *are prime.*

Suppose there are two powers that are only 1 apart:

$$C^\beta - B^\alpha = 1$$

Since one may prime factorize the exponents, $\beta = z_1^{k_1} z_2^{k_2} \dots$ and $\alpha = y_1^{m_1} y_2^{m_2} \dots$,

$$(C^{z_1^{k_1-1} z_2^{k_2} \cdots})^{z_1} - (B^{y_1^{m_1-1} y_2^{m_2} \cdots})^{y_1} = 1$$

so that there are also two powers with prime exponents that are only 1 apart. (For instance, suppose B^8 and C^{15} differ by 1, then $(B^4)^2$ and $(C^3)^5$ are powers with prime exponents that differ by 1.)

Date: 1 October 2022.

2010 *Mathematics Subject Classification.* Primary 11D41; Secondary 11A99.

This illustrates that starting from the assumption that the exponents y and z are prime, and arriving at the nonsolution of the Catalan equation (apart from a few exceptions); that such a proof by contradiction, would extend to all powers of B, C, and so show that there are no solutions, apart from those few exceptions. The few exceptions can be worked out, and they will turn out to include $2^3 + 1 = 3^2$. The rest will be seen to be trivial.

3. The Obvious Proof Using Calculus and Geometric Reasoning

Firstly, let's do this using calculus and geometric reasoning. Assume that exponents y and z are positive integer primes.

Proof. From equation (1), let θ be defined as the continuous variable

$$\theta = B^y = C^z - 1 \tag{2}$$

where $B = B(\theta)$ and $C = C(\theta)$ are continuous functions which are the coordinates of the point (x, θ) on their curves. The exponents y and z are to be regarded as constants, since they are positive integer primes.

Geometrically, for each level θ, there is one value of B, at the point (B, θ); and similarly, one value of C, at the point (C, θ). Inverting, one thus has

$$B(\theta) = \theta^{1/y} \qquad \text{and} \qquad C(\theta) = (\theta + 1)^{1/z}$$

For positive integer primes y and z, one may simply drop the root of unity.

Differentiating and keeping to real numbers,

$$B'(\theta) = \frac{1}{y}\theta^{1/y-1} \qquad \text{and} \qquad C'(\theta) = \frac{1}{z}(\theta + 1)^{1/z-1}$$

For positive odd primes y and z, clearly the root of unity must be positive.

If one increases $\theta \to +\infty$ without bound, then $B'(\theta) > 0$ and $C'(\theta) > 0$.

In the case of the even prime, this interval makes one choose the positive root of unity.

So $B(\theta)$ and $C(\theta)$ are monotonically increasing functions as $\theta \to +\infty$. They diverge. Which is to say, a solution to (2) is only possible for some low values of θ smaller than some threshold.

To be concrete, at the special level $\theta = \theta_0$, these will be two points $(C_0 = C(\theta_0), \theta_0)$ and $(B_0 = B(\theta_0), \theta_0)$ that obey the relation (2) above. This level is $\theta_0 = 8$, $B_0 = 2$, $C_0 = 3$, $z = 2$ and $y = 3$. One can see that $C_0 > B_0$ at this level θ_0. Higher than this level $\theta > \theta_0 = 8$, one has $1/z = 1/2 > 1/y = 1/3$. In this region, $\theta(C) > \theta(B)$ and $C > B$. Let's put the values $1/z = 1/2$ and $1/y = 1/3$ into the above, one has

$$B'(\theta) = \frac{1}{3}\theta^{1/3-1} = \frac{1}{3}\theta^{-2/3} \qquad \text{and} \qquad C'(\theta) = \frac{1}{2}(\theta + 1)^{1/2-1} = \frac{1}{2}(\theta + 1)^{-1/2}$$

So in this region $\theta > 8$, both $C'(\theta) > 0$ and $B'(\theta) > 0$. In other words, the functions are monotonically increasing with increasing θ. (But the derivatives B' drops to zero faster than C'.) Therefore as $\theta_0 < \theta \to +\infty$, $C(\theta) > B(\theta)$.

Which proves that $\theta(C_0) = \theta(B_0) = \theta_0$ is the only solution. ■

The objection to this proof is that it is not a pure number theory proof. That calculus should not be involved in such a proof about integer relations.

4. An Explicit Proof by Polynomial Factoring

Now an explicit derivation is presented using polynomial factoring. Assuming again that y and z are primes, the following can be claimed.

Claim 2. *y and z are different primes, i.e. $y \neq z$.*

Proof. Suppose $y = z$. Then (1) becomes, taking absolute value,

$$1 = |C^z - B^z| = |C - B| \left| \sum_{j=0}^{z-1} C^j B^{z-1-j} \right|$$

The first bracketed expression on the right hand side have an absolute value ≥ 1. The second is > 1 strictly. Therefore this cannot be equal. The supposition must be wrong, i.e. $y \neq z$. ■

Claim 3. *B and C are coprime, i.e. $gcd(B, C) = 1$.*

Proof. Suppose they are not. Then $\exists f, \hat{B}, \hat{C} \in \mathbb{Z}$ with $f > 1$ so that $B = f\hat{B}$ and $C = f\hat{C}$. Then equation (1) becomes, taking modulo f,

$$f^y \hat{B}^y = f^z \hat{C}^z - 1 \qquad \Longrightarrow \qquad 0 = -1 \pmod{f}$$

which is a contradiction. Hence B and C are coprime. ■

Claim 4. *B and C have opposite parity.*

Proof. The Catalan equation (1) has two terms B^y and C^z that have a difference of 1. Hence B^y and C^z and so B, C have opposite parity. ■

Claim 5. *Either $B < C$ or $C < B$.*

Proof. Since C and B are coprime, they are unequal: $B \neq C$. ■

There are therefore two cases to consider: $C < B$ and $B < C$. It will be seen that $C < B$ is impossible, while $B < C$ with B integer severely restricts the allowable solutions to $3^2 - 2^3 = 1$; (plus some other trivial solutions).

4.1. **Consider $C < B$.** Then write $B = q'C + r$ where $q', r \in \mathbb{Z}^+$ and $0 < r < C$. Factor the Catalan equation (1) this way

$$C^z = B^y + 1 = \underbrace{(B+1)}_{g(B)} \underbrace{(B^{y-1} - B^{y-2} + \cdots + 1)}_{h(B)} \tag{3}$$

Claim 6. *With $C < B$, $C \nmid g(B)h(B)$ whether C is composite or prime.*

Claim 7. *With $C < B$, C cannot be a prime.*

Proof. Assume for the sake of contradiction that C is prime. From equation (3) one may write

$$C^s = g(B) = B + 1$$
$$C^{z-s} = h(B) = B^{y-1} - B^{y-2} + \cdots + 1$$

Note that $s \geq 1$ and $z - s \geq 1$. Then $B = C^s - 1$ and one may substitute this into the expression for $h(B)$:

$$h(B) = (B^{y-1} - B^{y-2} + \cdots + 1) = \sum_{j=0}^{(y-1)} (-B)^j = \sum_{j=0}^{(y-1)} (1 - C^s)^j$$

Since $C \mid h(B)$, it must be that the 0th-order term divides C since every higher order term is divisible by C. To find the 0th-order term, observe that there are y

binomial terms, the 0th-order term of each being 1. So the 0th-order term is y, a prime. So $C = y$.

Now use this result in the Catalan equation (1) In this equation, B, C must have opposite parity. Since $C < B$, C must be the even term, and B the odd term. Then $C = y = 2$, the even prime and (1) becomes

$$\begin{aligned} & C^z = 2^z && = B^y + 1 = B^2 + 1 \\ \iff \quad & 2^z - 2 && = B^2 - 1 \qquad \text{(subtract 2 both sides)} \\ \iff \quad & 2(2^{z-1} - 1) && = (B-1)(B+1) \end{aligned}$$

On the left, since $z > 1$, the number is of the form $2(odd)$, a number with a single power of 2. On the right, both $(B-1)$, $(B+1)$ must be the same parity, so it is either odd or divisible by 4. This is a contradiction. ■

Now the case of C prime is proven. In Claims 8 and 10 below, one shall deal with C being composite. The two claims assume opposite parities: C even and B odd versus C odd and B even. This covers all the possibilities of C being composite, when $C < B$.

Claim 8. *With $C < B$, it cannot be that C is an even composite and B is odd.*

Proof. Firstly, assume for the sake of contradiction that C is composite. Factor $C = AE$ with $E > A \geq 1$ and $\gcd(A, E) = 1$. For complete generality, it is further assumed that both A and E are composite, although if it turns out not be, it is not a big deal. With these definitions, B is coprime to A, E. Then picking up from equation (3),

$$A^z = g(B) = B + 1 = q'C + r + 1 \tag{4}$$

$$E^z = h(B) = B^{y-1} - B^{y-2} + \cdots + 1 = \sum_{j=0}^{y-1} (-B)^j \tag{5}$$

Taking modulo A of the first equation (4), one may deduce that $r+1 \equiv 0 \pmod{A} \implies r = A - 1$. Then

$$B = A^z - 1 = (A-1)(A^{z-1} + A^{z-2} + \cdots + 1) = r\frac{A^z - 1}{A - 1}$$

and hence $r \mid B$. Since $B = q'C + r$, this implies that $q'C \equiv 0 \pmod{r}$ and therefore

$$q' = rs \quad \text{for some } s \in \mathbb{Z} \quad \implies B = q'C + r = r(sC + 1)$$

From equation (4), substituting $r + 1 = A$ and taking modulo a prime p_E that divides E, i.e. $p_E \mid E$,

$$A^z = q'C + r + 1 = q'AE + A \quad \implies \quad A^{z-1} \equiv 1 \pmod{p_E} \tag{6}$$

Since z is also a prime, it must be that $p_E = z$. (It is Fermat's Little Theorem for the prime z.) But this must hold for all primes $p_E \mid E$ even if repeated, so the conclusion is that $E = z^{\hat{k}}$ for $\hat{k} \in \mathbb{Z}$, $\hat{k} \geq 1$, constraining E to be a prime power, while A remains (so far) a composite. In other words, C factors as $C = Az^{\hat{k}}$, with $1 \leq A < E = z^{\hat{k}}$. Note that A must have primes that are distinct from z, and they may be repeated.

Now $p_E = z \neq 2$. Suppose for the sake of contradiction that $z = 2$. One has $A < E = z^{\hat{k}} = 2^{\hat{k}}$. Then $C = 2^{\hat{k}}A$. Put this result back into the Catalan equation (1):

$$C^z - 1 = (2^{\hat{k}}A)^2 - 1 = (2^{\hat{k}}A - 1)(2^{\hat{k}}A + 1) = B^y = F^y G^y$$

where again one factors B into the odd composite factors F and G, $F < G$, and $\gcd(F, G) = 1$ to correspond with factors $(2^{\hat{k}}A - 1)$ and $(2^{\hat{k}}A + 1)$.

$$\begin{cases} F^y &= 2^{\hat{k}}A - 1 \\ G^y &= 2^{\hat{k}}A + 1 \end{cases} \implies G^y = F^y + 2 \tag{7}$$

Therefore, one has two y^{th}-powers, a distance 2 apart.

Can one ever have two numbers, both y^{th}-powers, a distance 2 apart?

Claim 9. *Two p^{th} powers are always at least p apart.*

Proof. Consider the difference between a positive number N and $N + d$ both raised to the pth power, a distance $d > 1$ apart:

$$\begin{aligned} (N+d)^p - N^p &> (N+1)^p - N^p && (\because \text{ drop } d \text{ to } 1) \\ &= N^p + \binom{p}{1} N^{p-1} + \cdots - N^p && (\because \text{ binomial expansion}) \\ &> pN^{p-1} + \frac{p(p-1)}{2} N^{p-2} && (\because \text{ truncation of higher powers}) \\ &> pN^{p-1} && (\because \text{ truncation again}) \\ &> \min_{N \in \mathbb{Z}^+} (pN^{p-1}) = p(1^{p-1}) = p \end{aligned}$$

■

But one is assuming for the sake of contradiction that $z = 2$, so $p = y > 2$ is the other prime. Two y^{th} powers will never be distance 2 apart. So $p_E = z \neq 2$, neither p_E nor z are multiples of 2; they are odd primes. Equation (7) is not satisfiable for $z = 2$.

Continuing the proof, and picking up from equation (6), dividing throughout by A and bringing the $q'E$ term to the left,

$$A^z = q'AE + A \implies A^{z-1} - q'E = A^{z-1} - q'z^{\hat{k}} = 1$$

Now $z > 2$ and one may choose whether q' is odd or even.

$$q' = \begin{cases} odd & \iff q'z^{\hat{k}} \text{ is odd, } A \text{ therefore } C \text{ is even, } B \text{ is odd} \\ even & \iff q'z^{\hat{k}} \text{ is even, } A \text{ therefore } C \text{ is odd, } B \text{ is even} \end{cases} \tag{8}$$

This is the reason for the assumption of this claim (Claim 8). The proof of second case of even q' shall be deferred to that in Claim 10 below.

Now given that C is even and noting that $z > 2$, then $8 \mid C^z$. Taking modulo 8 of the Catalan equation (1), $0 \equiv B^y + 1 \pmod 8$. Since $(odd)^2 \equiv 1 \pmod 8$, y must be an odd prime, or else a contradiction arises.

Summarizing: At this point, one knows that both z and y are odd primes. $C = Az^{\hat{k}}$ with $\gcd(z^{\hat{k}}, A) = 1$ and A is even. B is also composite and odd, for complete generality.

Let p_A be an odd prime of A that is not z and not 2; i.e. $p_A \mid A, p_A \neq z, p_A \neq 2$. Then modulo p_A, the Catalan equation (1) becomes

$$B^y \equiv -1 \pmod{p_A} \implies B^{2y} \equiv 1 \pmod{p_A}$$

Therefore $\text{ord}_{p_A}(B) \mid 2y$ which means that

$$\text{ord}_{p_A}(B) = \begin{cases} 1 & \implies p_A = 2 \text{ contradiction! } \because p_A \neq 2 \\ 2 & \implies p_A = B + 1 = 3 \text{ contradiction! } \because B \text{ should be odd.} \\ y & \text{contradiction! } \because B^y \equiv -1 \pmod{p_A} \\ 2y & \implies p_A = 2y + 1 \end{cases}$$

Thus $p_A = 2y + 1$ is the only admissible conclusion from this.

Let p_B be an odd prime of B; i.e. $p_B \mid B$. Taking the Catalan equation (1) and modulo p_B, one has

$$C^z \equiv 1 \pmod{p_B}$$

And $\mathrm{ord}_{p_B}(C) \mid z$ which means that

$$\mathrm{ord}_{p_B}(C) = \begin{cases} 1 & \implies p_B = 2 \text{ contradiction! } \because B \text{ is odd.} \\ z & \implies z \mid p_B - 1 \implies p_B = 2mz + 1 \text{ for some } m \in \mathbb{Z} \end{cases}$$

Then $p_B = 2mz + 1$ is the only admissible conclusion from this.

Thus $C = 2 \cdot (2y + 1) \cdot z^{\hat{k}}$ and $B = 2mz + 1$. Substituting these values into the Catalan equation,

$$C^z - 1 = B^y$$

$$(2 \cdot (2y + 1) \cdot z^{\hat{k}})^z - 1 = (2mz + 1)^y \quad \implies \quad -1 \equiv 1 \pmod{z}$$

Now $z = 2$ is the only modulus where this is not a contradiction. But it has already been proven that $z \neq 2$. Thus assuming $C < B$ with C being even composite and B being odd, is not possible in this case. ■

This is the end of the proof of Claim 8. The following claim reverses the parities of C, B assumed above, and its proof shows that this too leads to a contradiction.

Claim 10. *With $C < B$, it cannot be that C is an odd composite and B is even.*

Proof. Now let's deal with q' even in equation (8). The proof is as above in the proof in Claim 8, as one repeats the steps gone through, noting the differences.

At the point of decision in equation (8), only z is known to be the odd prime. $C = Az^{\hat{k}}$ with $\gcd(z^{\hat{k}}, A) = 1$ and A is odd. B is also assumed composite and even, for complete generality.

Let p_B be a prime of B, i.e. $p_B \mid B$, remembering that B is even. Taking the Catalan equation (1) and modulo p_B,

$$C^z \equiv 1 \pmod{p_B}$$

And $\mathrm{ord}_{p_B}(C) \mid z$ which means that

$$\mathrm{ord}_{p_B}(C) = \begin{cases} 1 & \implies p_B = 2 \quad B = 2 \text{ strictly} \\ z & \implies z \mid \phi(B) \implies {}^{\dagger\dagger}B = 2mz^2 \text{ for some } m \in \mathbb{Z} \end{cases}$$

Consider the two cases above:

- The order of C in $\mathbb{Z}_{p_B}^{\times}$ must be divide $\phi(B)$ the totient function of B: $\mathrm{ord}_{p_B}(C) \mid \phi(B)$. If it is 1, then $p_B = 2$ and $B = 2$ is the only answer. This result is contradictory, because $C < B$ which means $C = 1$; C is also not composite. Then the Catalan equation (1) cannot be solved:

  $$C^z - 1 = 2^y \quad \iff \quad (1)^z - 1 = 2^y$$

  Thus when q' is even, the possiblity of $B = 2$ strictly is eliminated.
- ††Next consider $p_B = z$, which means that $z \mid \phi(B)$. Remember $2 \mid B$. Using the general result that $z^\alpha \mid B$ implies that $z^{\alpha-1}(z-1) \mid \phi(B)$, the smallest $\alpha = 2$ gives *one* prime factor $z \mid \phi(B)$. All of these considerations gives $B = 2mz^{\alpha \geq 2}$ for some $m \in \mathbb{Z}$. $\alpha \geq 2$ is the smallest possible power of z in B.

  Substituting these values for C and B into the Catalan equation,

  $$C^z - 1 = B^y$$

  $$(A \cdot z^{\hat{k}})^z - 1 = (2mz^{\alpha \geq 2})^y \quad \implies \quad -1 \equiv 0 \pmod{z}$$

which is a contradiction so long as $z > 1$. It may be observed that the exact value of A is not essential for making the contradiction manifest, as long as $C < B$; which can be done if the m, α in $B = 2mz^\alpha$ is *big enough.*

So finally: assuming $C < B$ and C is composite with C odd and B even is not possible. Claim 10 is proven. ■

In conclusion, assuming $C < B$, with C prime cannot satisfy the Catalan equation (1). It is the same with C composite, under consideration of all possibilities of B, C being odd or even.

4.2. **Consider $B < C$.** Then one may write $C = qB + r$ where $q, r \in \mathbb{Z}^+$ and $0 < r < B$, where q is the quotient and r the remainder.

Factor the Catalan (1) this way,

$$B^y = C^z - 1 = (C-1)(C^{z-1} + C^{z-2} + \cdots + 1) = \Phi_1(C)\Phi_z(C) \tag{9}$$

Use has been made of the fact that z is prime and thus factors into $\Phi_1(C)$ which is the first factor, and $\Phi_z(C)$ which is the second, a geometric series of C in z terms. (The use of cyclotomic polynomial labels $\Phi_1(C)$ and $\Phi_2(C)$ is just a fancy way of naming the polynomial. None of the deep theorems of cyclotomic polynomials are actually used.)

Claim 11. *If $r \neq 1$, then $B \nmid \Phi_1(C)$. Equation* (1) *is not satisfiable.*

Proof. The first factor is the polynomial $\Phi_1(C) = C - 1 = qB + r - 1$. If $r \neq 1$ and $r \neq 0$ still, then $B \nmid \Phi_1(C)$, so equation (9) has a factor that is not a factor of B^y, so equation (1) cannot be satisfied. ■

Claim 12. *If $r = 1$, then $B \mid \Phi_z(C)$ but $B^2 \nmid \Phi_z(C)$ (with exceptions). Equation* (1) *is not generally satisfiable.*

Proof. The second factor is the polynomial with prime index z,

$$\Phi_z(C) = C^{z-1} + C^{z-2} + \cdots + 1 = \frac{C^z - 1}{C - 1}$$

Substituting $C = qB + r$,

$$\begin{aligned}
\Phi_z(C) &= \frac{(qB+r)^z - 1}{qB + r - 1} \\
(qB + r - 1)\Phi_z(C) &= (qB+r)^z - 1 \\
&= \sum_{j=0}^{z} \binom{z}{j} (qB)^j r^{z-j} - 1 \\
&= \sum_{j=2}^{z} \binom{z}{j} (qB)^j r^{z-j} + \underbrace{(qB) z r^{z-1}}_{j=1} + \underbrace{r^z}_{j=0} - 1
\end{aligned}$$

With $r = 1$, one has, after cancelling a factor of (qB),

$$\Phi_z(C) = \sum_{j=2}^{z} \binom{z}{j} (qB)^{j-1} + z \tag{10}$$

So clearly, with $r = 1$, $B \mid \Phi_1(C)$ and $B \mid \Phi_z(C)$ iff $B \mid z$. Since z is prime, the last condition is simply $B = z$.

Now this condition may or may not lead to a contradiction. If B is composite, then a contradiction arises, and one is left with the conclusion that equation is not satisfiable, and hence the proof is complete. If however, B is prime, and then $B = z$ is possible.

Under this condition $B = z$, equation (10) becomes

(11)
$$\Phi_z(C) = B\left(\sum_{j=2}^{z}\binom{z}{j}q^{j-1}B^{j-2}+1\right) = B\left(\sum_{j=3}^{B}\binom{B}{j}q^{j-1}B^{j-2}+\underbrace{\binom{B}{2}q+1}_{H_z}\right)$$

where B has been factored out, and the constant term independent of powers of B is H_z. The factors in front are powers of B. But binomial coefficient $\binom{B}{2}$ need not be divisible by B because factors in the denominator may cancel factors of B in the numerator.

Clearly, $B \mid \Phi_z(C)$. Is $B^2 \mid \Phi_z(C)$? This depends on whether $B \mid H_z$.

- $B \nmid H_z$. In this case, the entire bracketed expression in equation (11) is not divisible by B, so in general, $B^2 \nmid \Phi_z(C)$. With one more factor of B in $\Phi_1(C)$, if $y > 2$ there are not enough factors to make B up to the yth power. So equation (1) cannot be satisfied.
- $B \mid H_z$. Then $Bm = H_z$ for some $m \in \mathbb{Z}$. Then one has this equation,

$$\begin{aligned} Bm = H_z = \binom{B}{2}q+1 &= \frac{1}{2}B(B-1)q+1 \\ \iff \qquad 0 &= B^2-(2m+q)B+2 \end{aligned}$$

This severely restricts the set of allowable B to those that satisfy the equation above. Solving for integer B using the quadratic root formula, one has only two solutions

$$2B_\pm = (2m+q) \pm \sqrt{\underbrace{(2m+q)^2-8}_{D^2}}$$

Noting the overall factor of 2, the right-hand side must be an *even* integer. To get integer solutions for $B_\pm$, one must let D^2 be an integer square and $2m+q \in \mathbb{Z}$. Noting the quantity under the square root, this means that 8 must be the difference between two squares:

$$(2m+q)^2 - D^2 = 8$$

But the squares are

$$1, 4, 9, 16, 25, \ldots$$

and therefore $D = 1^2$ and $(2m+q)^2 = 3^2$ is the *only* pair of squares that has a difference of 8. Only $2^2 - 1^2$ the other pair that is smaller. All other pairs give larger differences. So this is the only solution.

Substituting the two resulting roots $B_- = 1$ and $B_+ = 2$ into the Catalan equation (1),

- $B_- = 1$, $(B_-)^y + 1 = 1^y + 1 = 2^1$.
  While valid, this is discarded as a trivial solution since the powers must be > 1.
- $B_+ = 2$, $2^3 + 1 = 3^2$
  This is the *only* solution.

■

So the Catalan equation (1) also has no solution when $B < C$, apart from the trivial cases, and the only other one: $3^2 - 2^3 = 1$.

5. A Proof using Modular Arithmetic of a more general case

Consider the following equation, which is a generalization of the Catalan equation, where the equality has been replaced by the divides relation.

$$x^{(p-1)} \mid (p-1)^x + 1 \tag{12}$$

where p is a prime and x not necessarily so, but is an integer.

Clearly, any solution to the original Catalan equation (1) is a solution to (12), but not vice-versa. By finding the solutions to this equation, one may then check it against the original Catalan equation (1) to find if it was unique.

Proof. Equation (12) reduces to the following cases

- Let $x = 1$. Then (12) becomes

$$1^{p-1} \mid (p-1)^1 + 1 \qquad \implies 1 \mid p \tag{13}$$

  Hence $(x = 1, p)$ are all solutions to to (12).
- Let $x = 2$. Then (12) becomes

$$2^{p-1} \mid (p-1)^2 + 1 \tag{14}$$

  If $p = 2$, then the above equation becomes $2^1 \mid 2$. This means that$(x = 2, 2)$ is a solution.

  But if p is odd, then the above equation becomes

$$2^{\text{even}} \mid (\text{even})^2 + 1$$

  which is a contradiction.
- Now let $(x, p \geq 3)$, which means that p is an odd prime. Then equation (12) becomes

$$x^{\text{even}} \mid (\text{even})^x + 1 \qquad \implies x^{\text{even}} \mid \text{odd}$$

  Therefore x must be an odd integer in this case.

  Let $q \mid x$ be the smallest positive prime factor of x. Right at the outset, one knows that $q \neq 1$ and q is an odd prime. Which implies that

$$(p-1)^x + 1 \equiv 0 \pmod{q} \qquad \implies q \nmid (p-1)^x, \text{thus } q \nmid (p-1) \tag{15}$$

  Now Fermat's Little Theorem, applied to the nondivisibility of $p-1$ into q means that

$$(p-1)^{q-1} \equiv 1 \pmod{q} \tag{16}$$

  From $q \mid x$ being the smallest positive prime factor of x, one may deduce that any smaller number, which are

$$0 < 1, 2, \ldots, q-1 < q$$

  must be coprime to x. This means that $\gcd(x, q-1) = 1$. This implies that $\exists a, b \in \mathbb{Z}$, such that

$$ax - b(q-1) = 1$$

  which is simply Bezout's Identity.

  Notice from this that one of $b(q-1), ax$ is even the other is odd. Which one?

  One knows that $q-1$ must be even, since q itself is an odd prime, from the supposition that $q \mid x$ with x odd (i.e. it cannot be *even*). So $b(q-1)$ must the even term. To be consistent, ax is the odd term; knowing that x is odd, a must be odd. Notice that one still does not know if b is odd/even.

  Now take equation (16), raised to the bth power, and multiplying by $(p-1)$,

$$(p-1)[(p-1)^{q-1}]^b \equiv (p-1)^{1+b(q-1)} \equiv (p-1)^{ax} \equiv [(p-1)^x]^a \pmod{q}$$

The extreme left and right hand sides of the above becomes

$$(p-1) \equiv (-1)^a \pmod{q}$$

since $q \mid$ equation (12) and also (15). In other words, it is immaterial whether b is odd or even, by way of Fermat's Little Theorem.

Remembering that a is odd,

$$(p-1) \equiv -1 \pmod{q} \qquad \implies q \mid p$$

But since $q \neq 1, 2$ is a prime, that are no primes it can divide, unless $q = p$. Indeed it must be so.

Now that one knows that $(q = p) \mid x$, it may be written that $x = pm$ for some $m \in \mathbb{Z}$. Then (12) becomes, (remembering at p is an odd prime)

$$(pm)^{p-1} \mid (p-1)^{(pm)} + 1 \tag{17}$$

Setting $m = 1, 2, 3, \ldots$

$$\begin{aligned} p^{p-1} &\mid (p-1)^p + 1 \\ (2p)^{p-1} &\mid ((p-1)^p)^2 + 1 \\ (3p)^{p-1} &\mid ((p-1)^p)^3 + 1 \\ (4p)^{p-1} &\mid ((p-1)^p)^4 + 1 \\ &\vdots \end{aligned}$$

and so on.

Consider the equation (17) modulo p. This becomes

$$(p-1)^{pm} \equiv -1 \pmod{p} \qquad \implies \qquad (p-1)^m \equiv -1 \pmod{p}$$

where the implication comes from Fermat's Little Theorem, knowing that p is prime. Which means that m is constrained to be odd in equation (17). This is consistent with the condition that x is an odd integer, not necessarily prime, in this case.

Start with $m = 1$. Then $x = p$. Now one can substitute all these values for x, p back into (12).

$$\begin{aligned} p^{p-1} \mid (p-1)^p + 1 &= \sum_{i=0}^{p} \binom{p}{i} p^i (-1)^{p-i} + 1 \\ &= \sum_{i=2}^{p} \binom{p}{i} p^i (-1)^{p-i} + \underbrace{\binom{p}{1} p (-1)^{p-1}}_{=p^2} + \underbrace{\binom{p}{0} p^0 (-1)^p}_{=-1} + 1 \\ &= p^2 \left(1 + \sum_{i=2}^{p} \binom{p}{i} p^{i-2} (-1)^{p-i} \right) \end{aligned} \tag{18}$$

The term in brackets is clearly not divisible by p. So $p^{p-1} \mid p^2$ at most.

Then there are only three cases to consider from $m = 1$:

$$\begin{aligned} &p^{p-1} \mid p^0 \implies p^{p-1} \mid 1 \quad \text{(impossible)} \\ &p^{p-1} \mid p^1 \implies p^{p-1} \mid p \implies p = 2 \quad \text{(impossible)} \\ &p^{p-1} \mid p^2 \implies p^{p-1} \mid p^2 \implies p = 3 \quad \text{(possible)} \end{aligned}$$

Therefore the solution in this case is $(x = 3, p = 3)$.

Now generalize to odd m, so that $x = mp$. Repeat the above calculation by substituting back into (12).

$$\begin{aligned}
(p-1)^{mp} + 1 &= \sum_{j=0}^{mp} \binom{mp}{j} p^j (-1)^{mp-j} + 1 \\
&= \sum_{j=2}^{mp} \binom{mp}{j} p^j (-1)^{mp-i} + \underbrace{\binom{mp}{1} p(-1)^{mp-1}}_{=mp^2} + \underbrace{\binom{mp}{0} p^0 (-1)^{mp}}_{=-1} + 1 \\
&= p^2 \left(m + \sum_{j=2}^{mp} \binom{mp}{j} p^{j-2} (-1)^{pm-j} \right) \\
&= mp^2 \left(1 + \frac{1}{m} \sum_{j=2}^{mp} \binom{mp}{j} p^{j-2} (-1)^{pm-j} \right)
\end{aligned}$$

The summation in brackets is divisible by m because of a factor (mp) in the binomial coefficients. With an additional 1, the entire bracketed expression is not divisible by m or p. In conclusion for odd $m \geq 3$,

$$(mp)^{p-1} \mid (p-1)^{mp} + 1 \implies (mp)^{p-1} \mid mp^2 \text{ at most}$$

Therefore this becomes

$$(mp)^{p-1} \mid mp^2 \implies \begin{cases} 2m \mid 4m \text{ where } p = 2 \\ 9m \mid 9m \text{ where } p = 3 \\ mp^{p-1} \nmid mp^2 \text{ where } p > 3 \end{cases}$$

The solution $(x \geq 3, p = 2)$ is not possible since $x = mp$ is odd and $p = 2$ is even. So $(x \geq 3, p = 3)$ is the case which survives.

Summarizing, one has the following solutions to equation (12), the generalization of Catalan's equation

$$(x = 1, p) \quad (x = 2, p = 2) \quad (x = 3, p = 3) \quad (x \geq 3, p = 3)$$

Now take each one and try to see it it satisfies the original Catalan equation (1),

$$\begin{aligned}
(x = 1, p): &\quad 1^{p-1} = (p-1)^1 + 1 \quad \text{(no)} \\
(x = 2, p = 2): &\quad 2^{2-1} = 2 = (2-1)^2 + 1 = 2 \quad \text{(yes, but powers} = 1) \\
(x = 3, p = 3): &\quad 3^{3-1} = 3^2 = (3-1)^3 + 1 = 2^3 + 1 \quad \text{(yes)} \\
(x \geq 3, p = 3): &\quad x^{3-1} = x^2 = (3-1)^x + 1 = 2^x + 1 \quad \text{(as above)}
\end{aligned}$$

The solution $(x = 2, p = 2)$ satisfies, but is unacceptable since it is demanded that the powers > 1 in equation (1). And the fourth case, one has the following equation

$$x^2 - 1 = 2^x \quad \iff \quad (x-1)(x+1) = 2^x$$

This reduces to

$$x - 1 = 2^k \text{ and } x + 1 = 2^{x-k} \quad \text{for some } k \in \mathbb{Z}$$

There is only one solutions to this,

$$\begin{aligned}
2 = 2^k(2^{x-2k} - 1) \qquad &\implies k = 1, x = 3 \\
2x = 2^k(2^{x-2k} + 1) \qquad &\implies k = 1, x = 2
\end{aligned}$$

with the second one $x = 2$ eliminated since it fails to satisfy $x^2 - 1 = 2^x$. In other words, the solution $(x \geq 3, p = 3)$ reduces to $(x = 3, p = 3)$.

From this, one can see that the only *unique* solution to Catalan's equation is $2^3 = 3^2 - 1$. ■

5.1. **Reflection.** By re-deriving the unique solution, without the addition of any others, under these less restrictive equation, shows quite convincingly that this is the only solution. Quite obviously, there will be none further on moving to even more relaxed constraints.

It must be stressed that while this is an inductive proof, it is not lacking in rigor. From the manipulations in equation (18), one can see that the analogous one of equation (11) of Section 4 is repeated. This later proof thus subsumes the first.

I open this third proof up for critique. Even if not successful, the first two proofs still hold up better than the *textbook* [2], right?

6. Conclusion

I have managed to proof Catalan's Conjecture three times: once using calculus and geometric reasoning; then a second time using by factoring and reasoning carefully about orders of integers in multiplicative fields modulo primes; and finally a third time by solving a more general problem: $n^{p-1} \mid (p-1)^n + 1$ using purely integer arguments. In the third time, a relaxation of the constraints of the original problem was performed to see if more solutions are generated.

7. Acknowledgements

I thank Chan Onn, Ph.D Mathematics (John Hopkins) for a critical review of this paper. He pointed out a critical error which has been corrected. Nevertheless, all errors are mine.

References

[1] Preda Mihăilescu. Primary cyclotomic units and a proof of catalan's conjecture. *J. Reine Angew. Math.*, 572:167-195, 2004.

[2] Rene Schoof. *Catalan's Conjecture*. Universitext. Springer, 2009.